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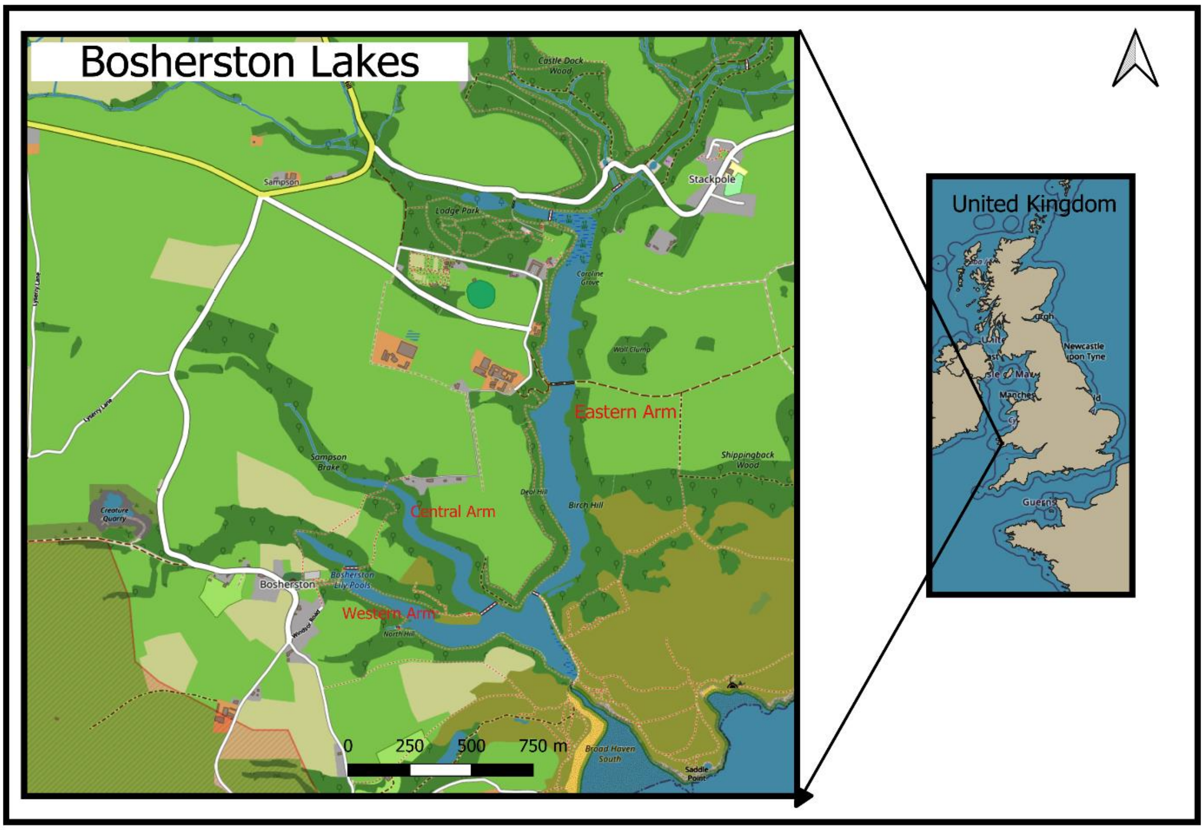
Water Chemistry Assessment: pH, Conductivity and Temperature

Background



A detailed map of Bosherston Lakes, located in Pembrokeshire, Wales, highlighting the key areas of the Eastern Arm, Central Arm, and Western Arm, with an inset map showing its location within the United Kingdom

The Bosherston Lily Ponds are a series of shallow, interconnected artificial lakes located within the Stackpole Estate in the Pembrokeshire Coast National Park, southwest Wales. The ponds were originally constructed between the late 18th and mid-19th century by the Cawdor family, who dammed the Bosherston stream to create ornamental lakes characteristic of Romantic landscape design. Over time, white water lilies (Nymphaea alba) were introduced and spread extensively, giving the ponds their distinctive ecological identity. Today, the site is owned and managed by the National Trust and designated as both a Site of Special Scientific Interest (SSSI) under the Wildlife and Country side Act and a Ramsar Wetland of International Importance due to its biodiversity, including notable aquatic macrophytes, invertebrates, breeding birds, and otter populations.

Hydrologically, the pond system is primarily fed by groundwater and stream inflow from a limestone catchment, which has important implications for water chemistry. The underlying carbonate geology produces naturally alkaline water with stable pH, while the ponds' shallow depth and slow water movement make them sensitive to seasonal temperature changes, rainfall-driven dilution, and evaporation effects.

Water flow typically moves from the inflow stream through the series of ponds and eventually toward the outflow at Broad Haven Beach, although this may be restricted during periods of low water level.

In this study, three water-quality indicators are examined: water temperature, pH, and conductivity. Water temperature is a key driver of ecological processes because it influences metabolic rates, dissolved oxygen availability, and the seasonal activity of aquatic organisms. pH reflects the balance between photosynthesis, respiration, and the buffering capacity of the underlying limestone; it can therefore vary with both biological productivity and hydrological inputs. Conductivity is used as a measure of the concentration of dissolved ions, and is influenced by groundwater chemistry, rainfall dilution, and evaporation concentration effects.

The ponds are subject to both seasonal variations (e.g., warmer summer temperatures, greater biological activity, and lower flow conditions) and long-term influences, such as climate warming, changes in land use within the catchment, and conservation management practices. For example, summer periods are typically associated with higher water temperatures and increased photosynthesis, which may result in higher pH during daylight due to carbon dioxide uptake. Conversely, winter rainfall tends to dilute ionic concentrations, leading to lower conductivity values and more stable pH. Over longer timescales, changes in rainfall patterns, drought frequency, and vegetation cover may affect how these variables fluctuate and interact.

Understanding how water temperature, pH, and conductivity vary seasonally and over multiple years provides insight into the physical, chemical, and biological functioning of the Bosherston Lily Ponds system. This information is crucial for the effective management and conservation of the ponds, particularly in the context of climate change, visitor pressure, and the protection of sensitive aquatic habitats.

Figures

1.1 Long Term Analysis.

Water Temperature in a long-term variation.

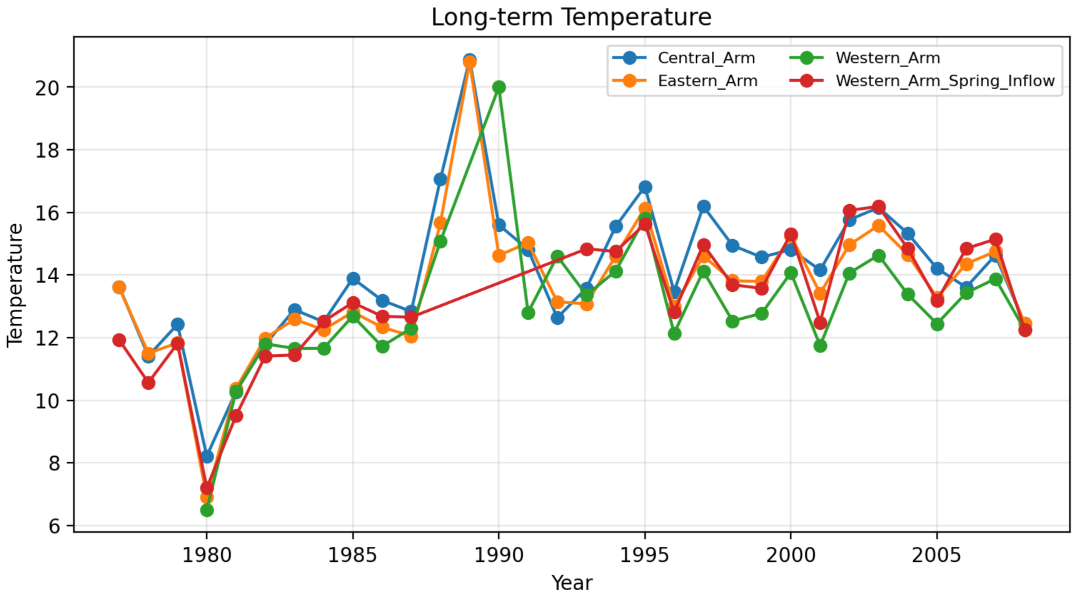


Figure 1. *Long-term trends in water temperature at Bosherston Lily Ponds show distinct multi-year fluctuations rather than a consistent warming trajectory. A marked decline occurred around 1980, followed by a rapid increase by 1983, and a short period of relative stability between 1983 and 1984. A pronounced warming event around 1986 raised temperatures to 20°C, twice the levels recorded in 1980. After 1990, temperatures decreased slightly and then stabilised into an oscillating regime of approximately 19°C between 1992 and 2006.*

These long-term variations reflect broader atmospheric warming patterns. Climate change has not only increased global near-surface temperatures but also altered regional climate regulators such as humidity, solar radiation and atmospheric circulation (Willett et al., 2020). The pattern observed at Bosherston is consistent with global lake-warming trends, where surface waters warmed by an average of 0.34°C per decade between 1985 and 2009 (O'Reilly et al., 2015). Further projections suggest an additional 1–4°C warming in lake systems over the next century (Woolway & Maberly, 2020; Grant et al., 2021).

These temperature increases have important ecological implications. Warmer waters hold less dissolved oxygen, increasing physiological stress on aquatic organisms, altering metabolic rates, and shifting species composition (Kraemer et al., 2021). Thus, the thermal changes observed in the Bosherston Lily Ponds may signal increasing vulnerability of the ecosystem to climate-driven stress.

Water pH in a long-term variation.

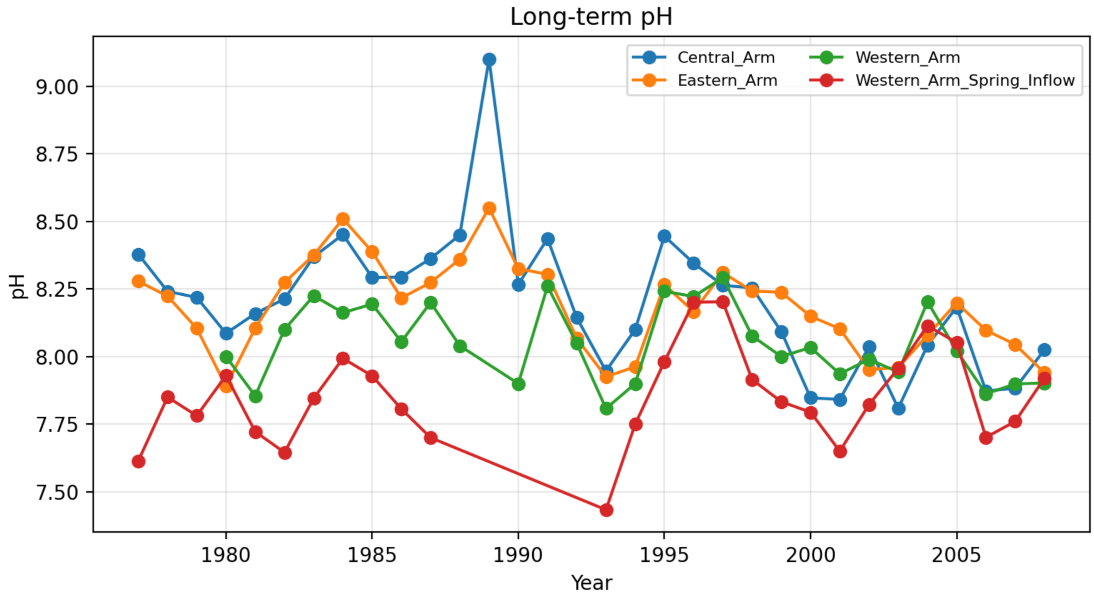


Figure 2. *Long-term variation in pH across the four ponds of the Bosherston Lily Ponds system from 1975 to 2007. The Central, Eastern, and Western Arms show similar temporal patterns with pH increasing through the late 1980s before stabilising and gradually declining in the early 2000s. In contrast, the Western Arm Spring Inflow consistently exhibits lower pH values and a more pronounced decline during the late 1980s, indicating external catchment influence on water chemistry. These patterns suggest that internal biological processes regulate pH in the main ponds, while nutrient-enriched inflow from the catchment drives a divergent chemical response in the Western Arm.*

The elevated pH observed in the late 1980s corresponds closely with a period of increased phosphorus enrichment in the Bosherston system. Phosphorus stimulated phytoplankton blooms, which enhanced photosynthetic uptake of CO_2 from the water column. The reduction in dissolved CO_2 decreased carbonic acid concentrations and therefore increased pH. During this period, the *Chara* beds in the Central Arm declined and subsequently disappeared. *Chara* requires clear, low-nutrient water and is highly sensitive to reductions in light availability; therefore, increased phytoplankton biomass and reduced water transparency caused by eutrophication likely suppressed *Chara* growth. The loss of *Chara* further reduced the system's buffering capacity and shifted the pond towards a more phytoplankton-dominated state. Thus, phosphorus enrichment in the 1980s directly links to both the elevated pH and the collapse of *Chara* beds. This is consistent with (Kosten et al., 2009) who found that at a global scale submerged macrophytes clearly decreased with increase in nutrients which results in increased algae biomass which reduces the amount of light that is needed for the survival of *Chara.* A correlation existence between light and distribution of macrophytes has been established (Spence and Chrystal, 1970).

Water Conductivity in a long-term variation.

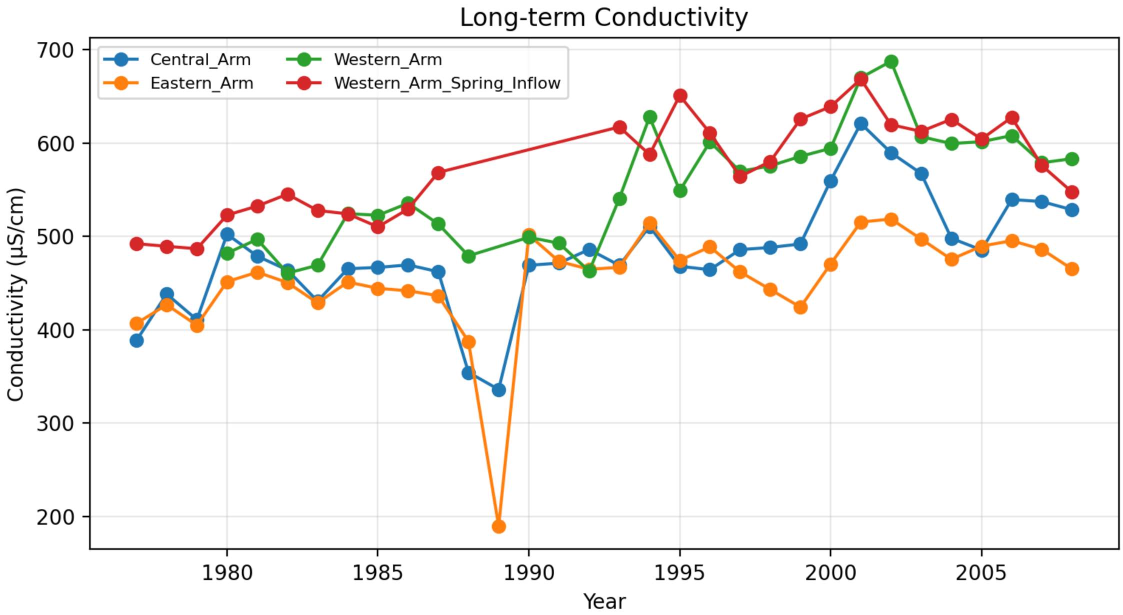


Figure 3. *Long-term variation in conductivity across the four ponds of the Bosherston Lily Ponds system (1975–2007). Conductivity increases over time in all ponds, with the Western Arm Spring Inflow consistently showing the highest conductivity, indicating sustained input of dissolved ions from the surrounding catchment. The Western Arm also displays elevated conductivity relative to the Central and Eastern Arms, reflecting its hydrological connection to the inflow. Periods of sharp fluctuation, particularly around the late 1980s and early 1990s, correspond to episodes of nutrient enrichment and changing land-use practices in the catchment. Overall, the patterns suggest that external catchment-derived inputs play a significant role in shaping water chemistry, especially in the Western Arm system.*

During the late-1980s, conductivity declined sharply across the pond arms while pH and temperature rose, whereas the Western Arm spring inflow showed a slight conductivity increase. The divergent behaviour is consistent with internal carbonate chemistry in the ponds versus external ionic loading at the inflow. Elevated temperature and productivity increased pH via CO_2 uptake, promoting calcite precipitation and thereby lowering conductivity in the ponds. In contrast, the inflow's conductivity rise coincides with agricultural runoff (evidenced by elevated K^+), indicating higher external ion inputs. Windowed analyses (1985–1990) show a strong negative association between pond pH and pond conductivity, and a positive association between inflow K^+ and inflow conductivity, while paired regressions confirm that pond conductivity decreases with increasing pond pH even after accounting for inflow conductivity. Together these results support a mechanism of catchment-derived ionic inputs at the inflow coupled with in-pond biogeochemical removal of ions during warm, productive years. The divergent behaviour is associated with long-term trends in lake temperature that are mainly controlled by regional climate warming.
The short-term shifts in pH and conductivity, especially in the 1980s, are driven by anthropogenic nutrient inputs from surrounding farmland, which altered biological activity and internal water chemistry. Rodriguez et al. (2018) also reported that biological processes, particularly photosynthetic uptake of CO_2, exerted a strong control on pH gradients within shallow lake systems.

Their findings demonstrated that internal biogeochemical processes could interact with external drivers such as climate variability and catchment runoff, resulting in diverging patterns in water chemistry over time. Trends in P flux may be driven by several factors including changes in fertilizer or manure use, management practices intended to reduce P transport, total seasonal or annual precipitation, and the frequency and magnitude of extreme precipitation events (Garnache et al. 2016; Gillon et al. 2016).

1.2 Seasonal Term Analysis

Water Temperature in annual variation

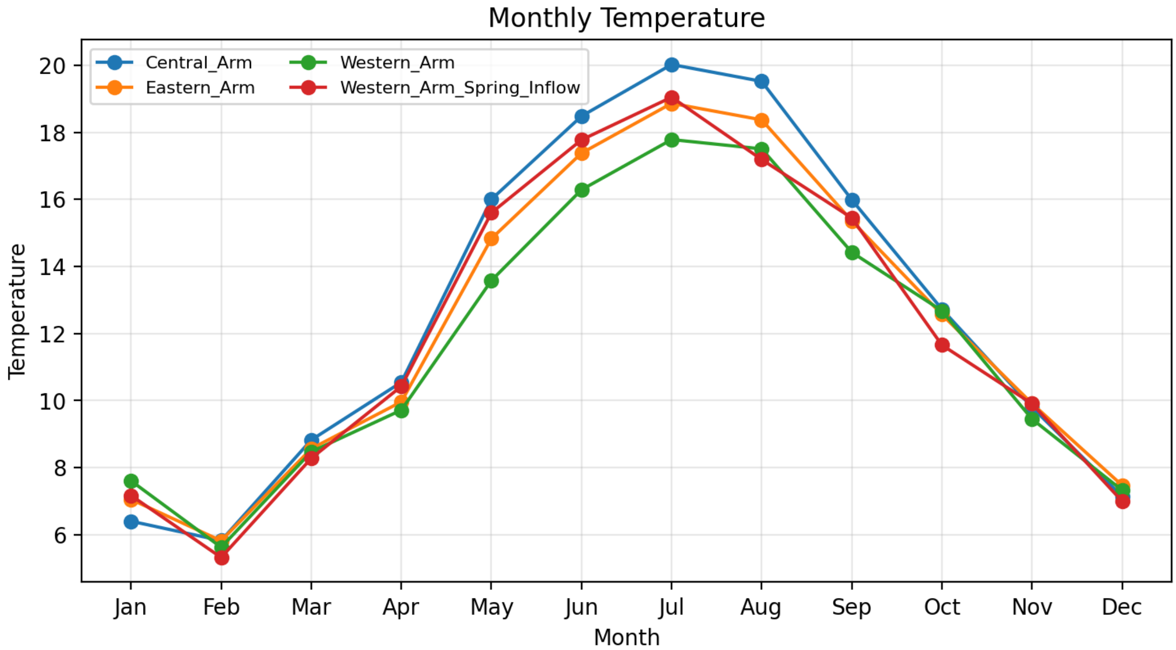


Figure 4. *Seasonal patterns in water temperature were consistent across all four ponds. Temperatures increased progressively towards the summer months, reaching their highest values during mid-summer, and declined during the winter period. A secondary decrease was also evident during the autumn transition period.*

The fitted quadratic regression model indicated a strong relationship between atmospheric (air) temperature and water temperature, with an R^2 value of 0.71, suggesting that approximately 71% of the seasonal variability in water temperature can be explained by changes in air temperature. This demonstrates that the ponds respond sensitively to seasonal climatic forcing, particularly solar radiation and ambient temperature conditions. This strong climatic coupling is typical of shallow lakes and ponds, where limited thermal buffering allows water temperature to track air temperature closely.

Water Conductivity in annual variation

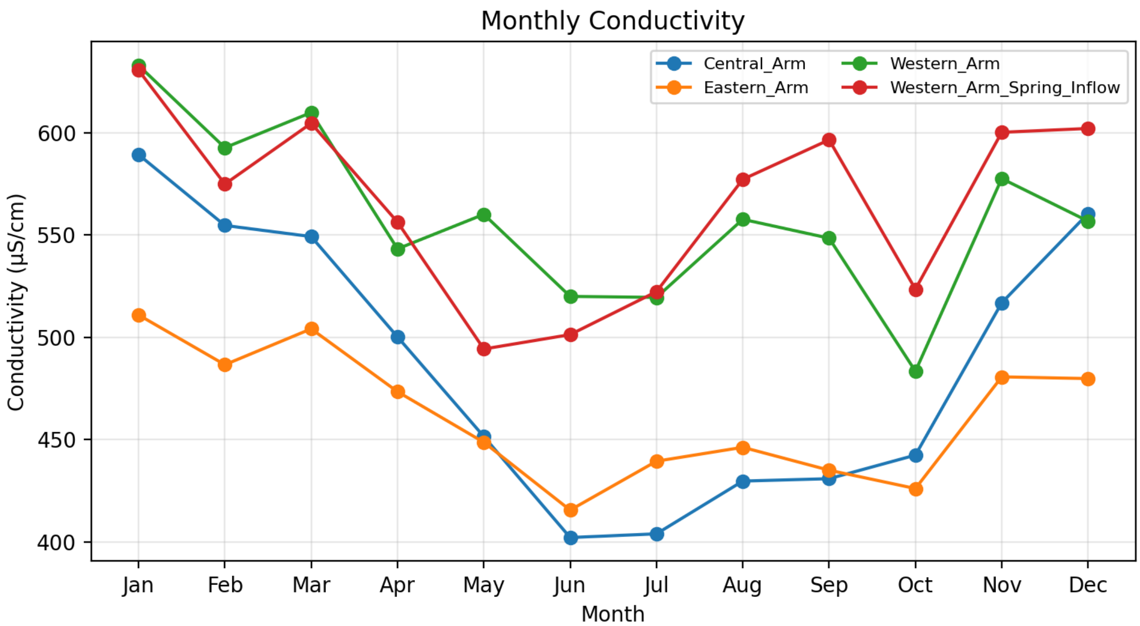


Figure 5. *Seasonal patterns in conductivity were consistent across most of the ponds, with lower conductivity during summer and higher conductivity during winter. This pattern reflects typical dilution during periods of greater biological activity and rainfall, and concentration of ions during colder, lower-activity months. However, the Western Arm Spring Inflow and the Western Arm displayed opposing behaviour, showing elevated conductivity during summer, creating a divergent chemical response relative to the other ponds.*

This divergence can be explained by external nutrient and ion inputs from the surrounding agricultural catchment. Runoff from farmland during summer rainfall delivers fertiliser-derived ions (e.g., K^+, NO_3^-, PO_4^{3-}) into the Western Arm Spring Inflow, raising its conductivity. Because the Western Arm receives continuous inflow from this source, its conductivity also increases, overriding the typical seasonal dilution and biological uptake effects observed in the other ponds.

Thus, while the other ponds are primarily influenced by internal biological and seasonal climatic processes, the Western Arm system is additionally shaped by short-term anthropogenic nutrient loading, producing an atypical conductivity signature during summer. Regression analysis shows a very strong relationship between conductivity in the western arm and conductivity in the western arm spring inflow.

Water pH in annual variation

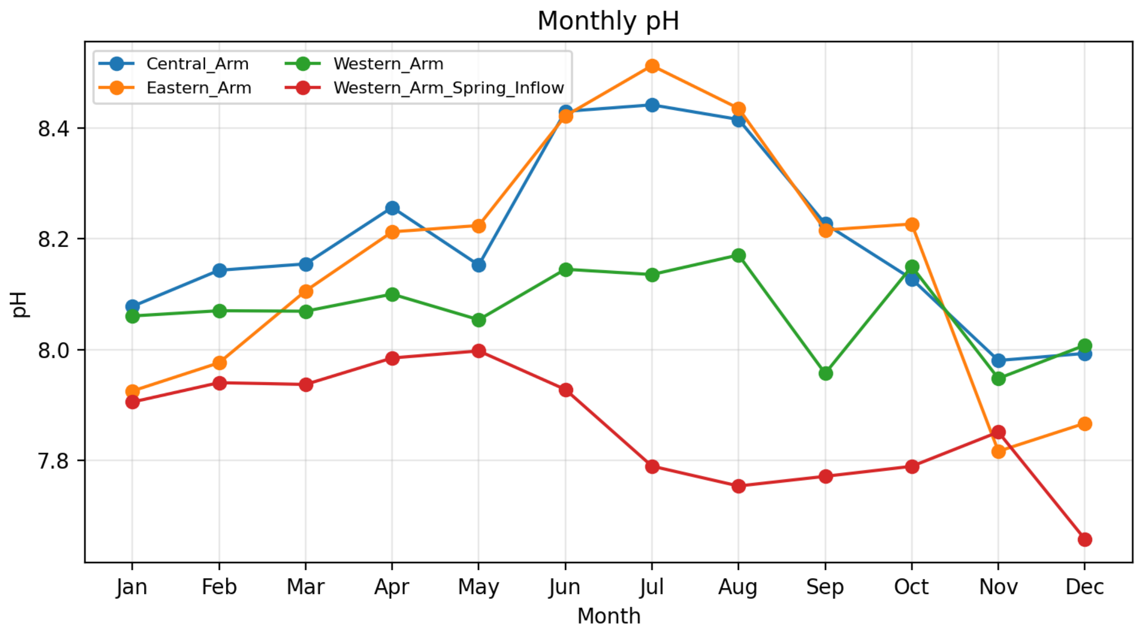


Figure 6. *Seasonal patterns in pH show a clear increase during summer in the Central, Eastern and Western ponds. This seasonal rise is driven by increased photosynthetic activity during warmer months, which reduces dissolved CO_2 and raises pH through a shift in the carbonate equilibrium. In contrast, the Western Arm Spring Inflow shows a decline in pH during summer, producing a divergent seasonal response.*

This suggests that the inflow is being influenced by catchment-derived inputs, such as nutrient-rich agricultural runoff, which increases microbial respiration and organic matter breakdown, generating CO_2 and lowering pH.

A sharp decrease in pH in the Western Arm in early autumn (September) aligns with this pattern. Because the Western Arm receives continuous inflow from the Western Arm Spring Inflow, the lower-pH water entering the system during late summer and early autumn appears to override the internal photosynthesis-driven pH elevation, leading to the observed drop. This indicates that external catchment processes can modulate, and at times dominating, internal biogeochemical dynamics in the Western Arm.

<u>**Statistical Operations**</u>

Kruskal–Wallis tests across sites

	Variable	H	p-value
0	Temperature	5.0428	0.1687
1	pH	121.6803	0.00000000000000000000000003
2	Conductivity	334.2315	0.000000000000000000000000000000000000000000000000000000000000000000000003

Figure 7. *The Kruskal–Wallis test indicated that water temperature did not differ significantly among the four ponds, suggesting that thermal conditions are relatively uniform across the system, due to shared climatic forcing and hydraulic connectivity. In contrast, pH and conductivity differed significantly among ponds ($p < 0.05$), indicating spatial variation in water chemistry within the system. This suggests that biogeochemical processes and catchment inputs influence chemical characteristics differently across the ponds.*

Significant differences in water conductivity among four arms

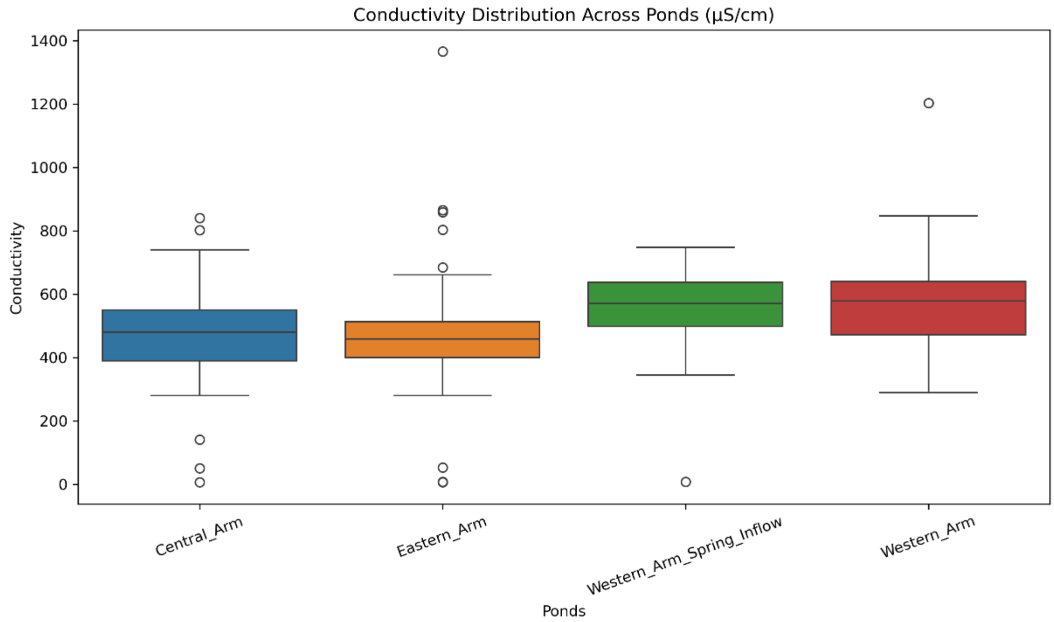


Figure 8. *Boxplot showing the distribution of conductivity (µS/cm) across the four ponds of the Bosherston Lily Ponds system. The Western Arm Spring Inflow and Western Arm exhibit consistently higher conductivity values compared to the Central and Eastern Arms, indicating greater ion and nutrient loading from catchment sources. The wider spread and higher upper ranges in these two sites reflect greater variability and episodic runoff-driven enrichment, whereas the Central and Eastern ponds display lower and more stable conductivity, suggesting they are influenced primarily by internal biological processes rather than direct external inputs.*

Significant differences in water pH among four arms

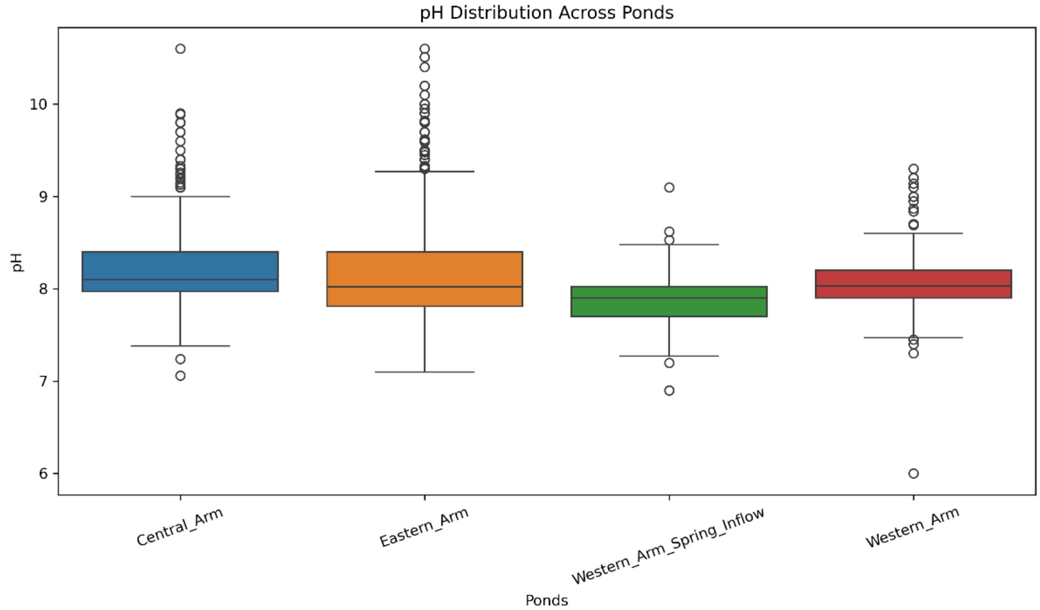


Figure 9. *Boxplot showing pH distributions across the four ponds of the Bosherston Lily Ponds system. The Central and Eastern Arms exhibit higher and more variable pH values, reflecting periods of enhanced photosynthetic activity and CO_2 drawdown. In contrast, the Western Arm Spring Inflow displays lower median pH and reduced variability, indicating greater influence from catchment-derived inputs and reduced carbonate buffering. The Western Arm shows intermediate values, consistent with its mixing position between externally influenced inflow and the internally regulated main ponds. These differences align with the significant spatial variation in pH identified in the Kruskal–Wallis test.*

Water Temperature insignificance difference

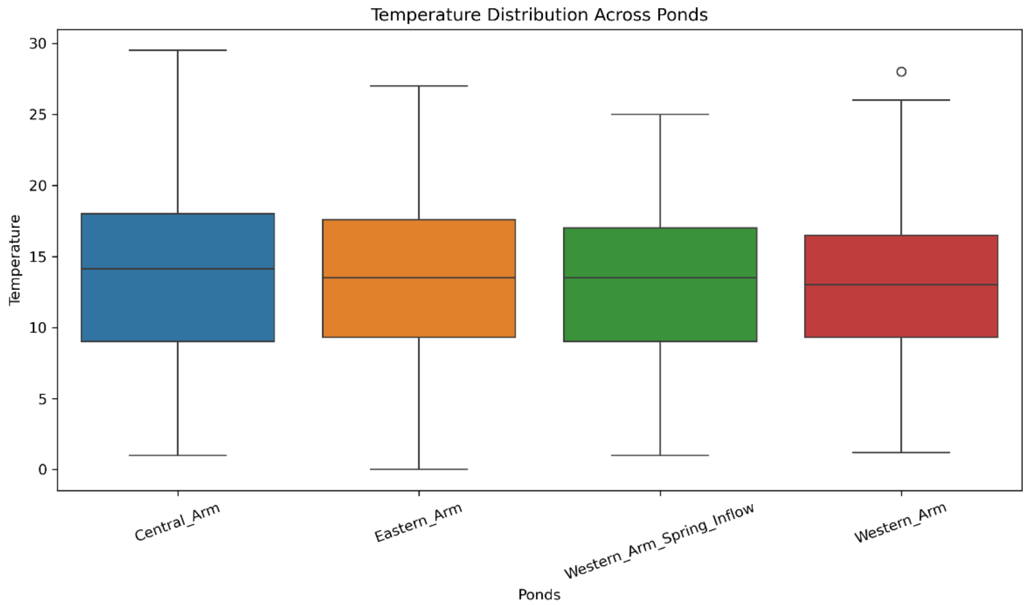


Figure 10. *Boxplot showing the distribution of water temperature across the four ponds of the Bosherston Lily Ponds system. All ponds display similar temperature ranges and median values, indicating that thermal conditions are uniform across the system. This reflects the strong influence of regional atmospheric temperature and shared hydrological connectivity, rather than localised internal processes. The absence of significant spatial differences in temperature is consistent with the Kruskal–Wallis test results, which showed no statistically significant variation in water temperature between ponds.*

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Correlations Statistics.

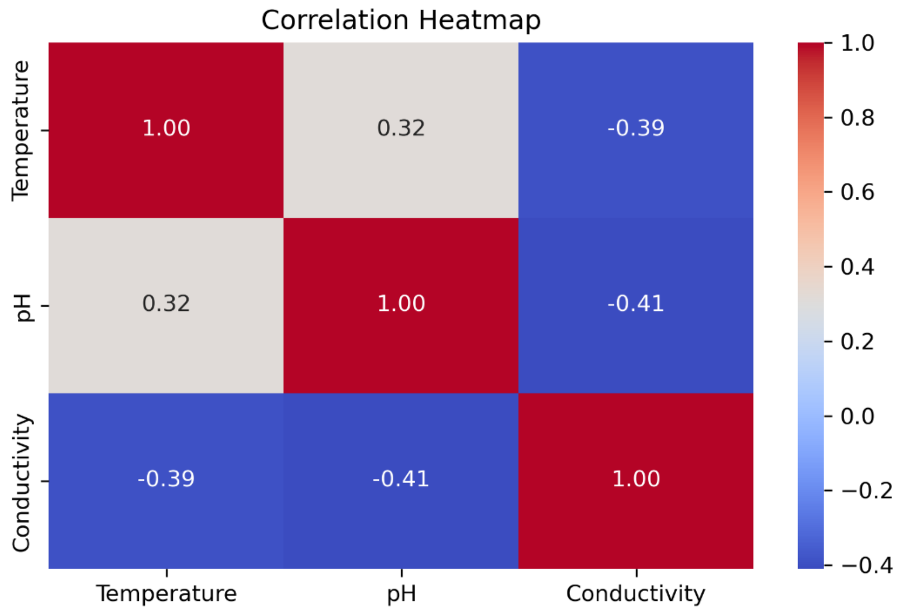

Figure 11. *The correlation heat map showed clear internal relationships among the water chemistry variables. Water temperature and pH were positively correlated (r = +0.32), indicating that pH tends to increase under warmer seasonal conditions, likely due to enhanced photosynthetic* CO_2 *uptake. In contrast, temperature and conductivity were negatively correlated (r = –0.39), suggesting that warmer periods are associated with biological dilution or carbonate precipitation processes that reduce dissolved ion concentration. Similarly, pH and conductivity were negatively correlated (r = –0.41), reflecting the role of high pH in promoting calcite precipitation and ion removal from the water column.*

Scatter Plot: Water conductivity and temperature.

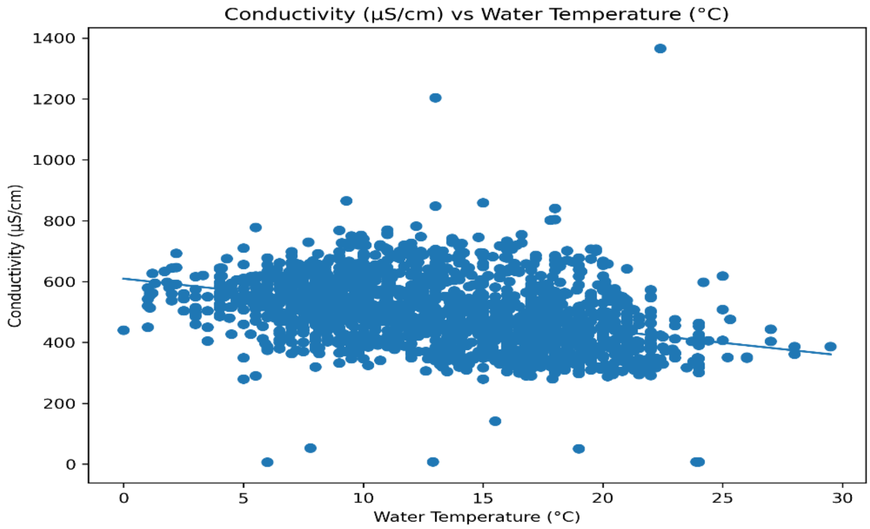

Figure 12. *Scatterplot showing the relationship between water temperature (°C) and conductivity (µS/cm) across the Bosherston Lily Ponds system. The data reveal a weak negative association, where conductivity tends to decrease as temperature increases. This pattern is consistent with temperature-driven biological processes, where warmer conditions enhance photosynthetic* CO_2 *uptake, shifting the carbonate equilibrium and promoting precipitation of calcium carbonate, thereby reducing dissolved ionic concentration. The broad spread of data points reflects variability across seasons and pond locations, but the overall trend supports the observed inverse relationship between temperature and conductivity.*

Scatter Plot: Water pH and temperature.

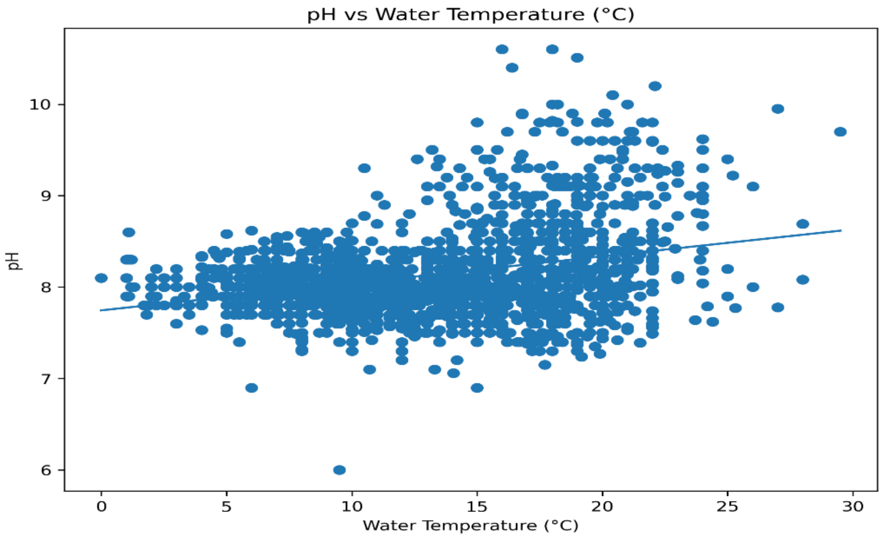

Figure 13. *Scatterplot showing the relationship between water temperature (°C) and pH across the Bosherston Lily Ponds system. The plot shows a weak positive association, with pH values tending to increase slightly as water temperature increases. This pattern is consistent with temperature-enhanced photosynthetic activity, where greater CO_2 uptake at warmer temperatures shifts the carbonate equilibrium and leads to higher pH. The wide spread of data points reflects seasonal and spatial variability across the ponds, yet the overall trend supports the observed temperature–pH coupling identified in the statistical and model-based analyses.*

Scatter Plot: Water conductivity and pH.

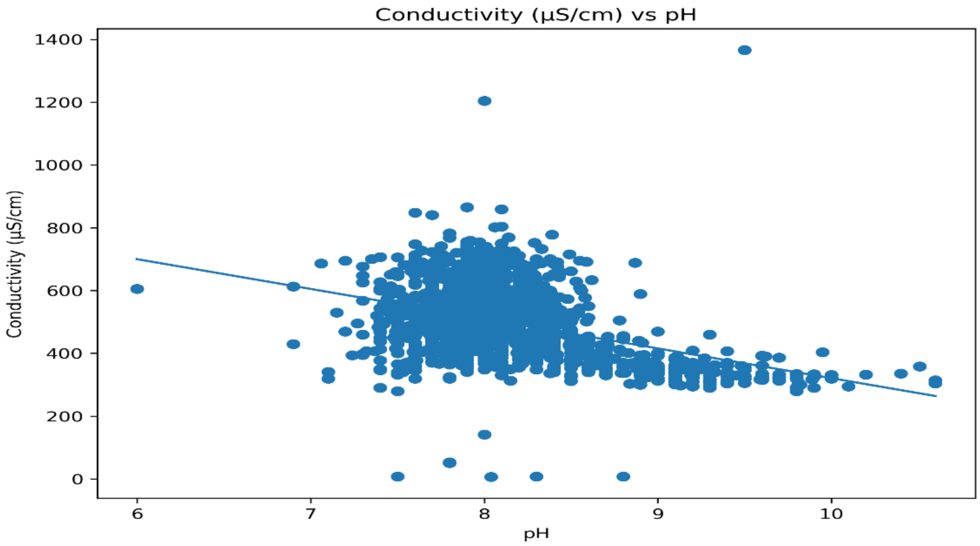

Figure 14. *Scatterplot showing the relationship between pH and conductivity (µS/cm) across the Bosherston Lily Ponds system. The plot indicates a negative association, with conductivity decreasing as pH increases. This pattern is consistent with photosynthesis-driven shifts in carbonate chemistry, where increased* CO_2 *uptake at higher pH promotes precipitation of calcium carbonate, thereby reducing the concentration of dissolved ions and lowering conductivity. The spread of data reflects spatial and temporal variation across ponds, but the overall trend supports the inverse conductivity–pH relationship observed in the correlation and model outputs.*

WESTERN ARM VS WESTERN INFLOW RELATIONSHIP

Linear Regression analysis

	Relationship	slope	intercept	R2	r	p
0	Western Arm Temperature (°C) ~ Inflow Temperature (°C)	0.8553	1.2768	0.8857	0.9411	0000000000000000000000000000000000000000000000000000000000000000000000000002
1	Western Arm pH ~ Inflow pH	0.4135	4.8194	0.1565	0.3956	0.00000005
2	Western Arm Conductivity (µS/cm) ~ Inflow Conductivity (µS/cm)	0.5576	239.3743	0.3192	0.565	0.0000000000000002

Figure 15. *There were significant relationships between the water chemistry of the Western Arm and the Western Arm Spring Inflow, indicating that the inflow exerts a measurable influence on the chemical conditions within the Western Arm. The linear regression results (see Table above) show that variations in pH, conductivity, and temperature in the Western Arm are partly explained by corresponding changes in the inflow. This suggests that catchment-derived inputs from the inflow propagate downstream, shaping the chemical profile of the Western Arm more strongly than in the other ponds.*

Western Arm Conductivity and Western Inflow Conductivity Relationship

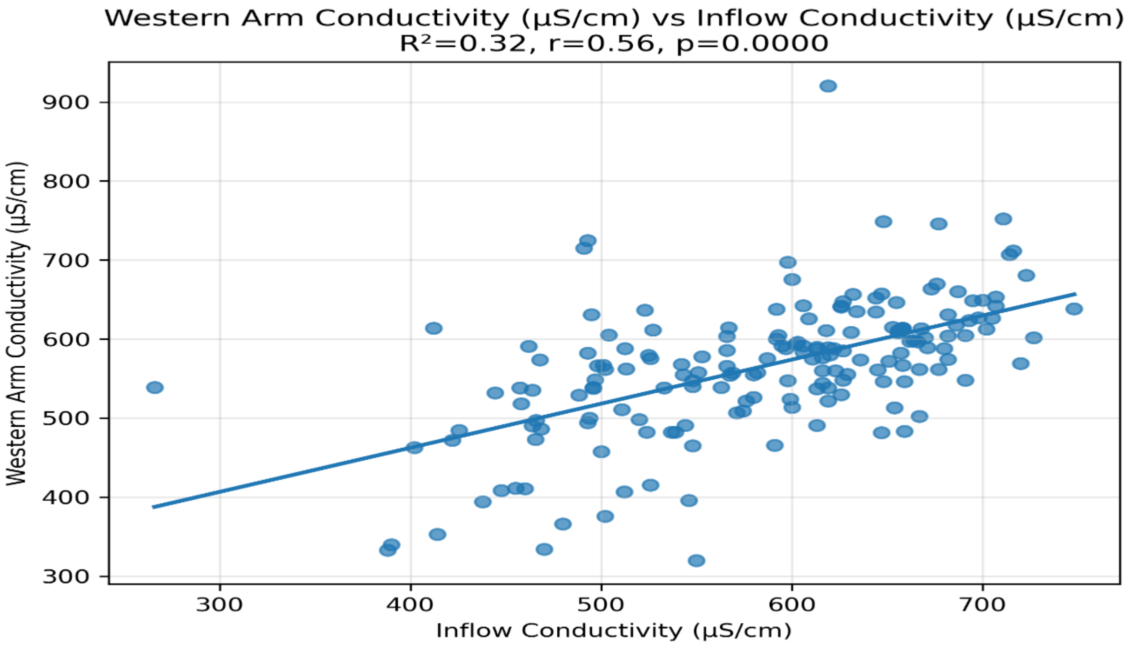

Figure 16. *Scatter Regression plot showing the relationship between the inflow water conductivity and Western Arm water conductivity. A positive relationship between water conductivity in the western spring inflow and the western arm.*

Western Arm pH and Western Inflow pH Relationship

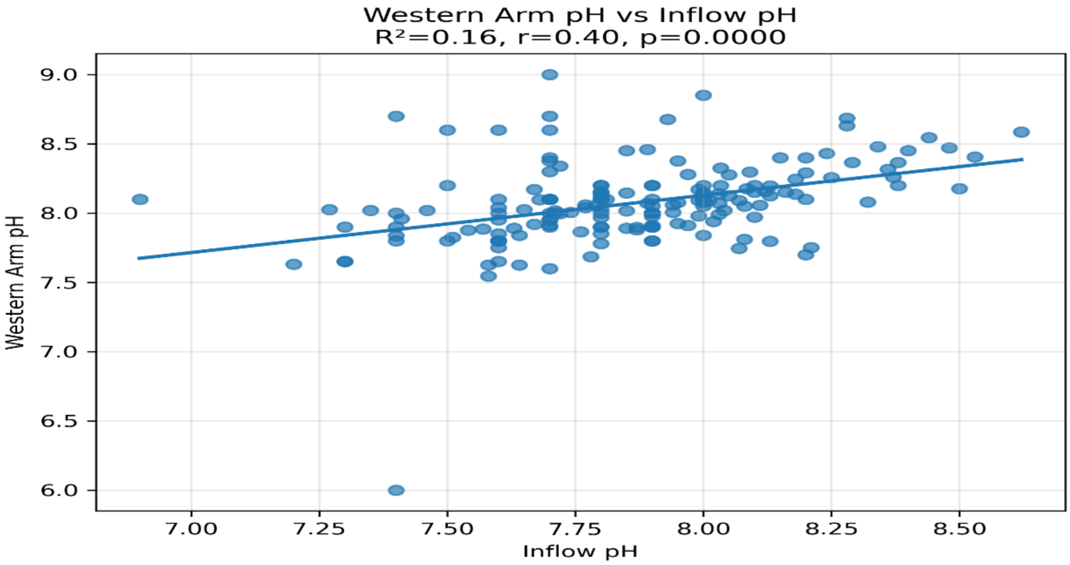

Figure 17. *A scatter plot and a regression line showing a positive relationship between water pH in the western arm and the water pH in the western spring inflow. There is a positive relationship between the Western Arm pH and the Western Spring Inflow.*

Western Arm Water Temperature and Western Inflow Temperature Relationship

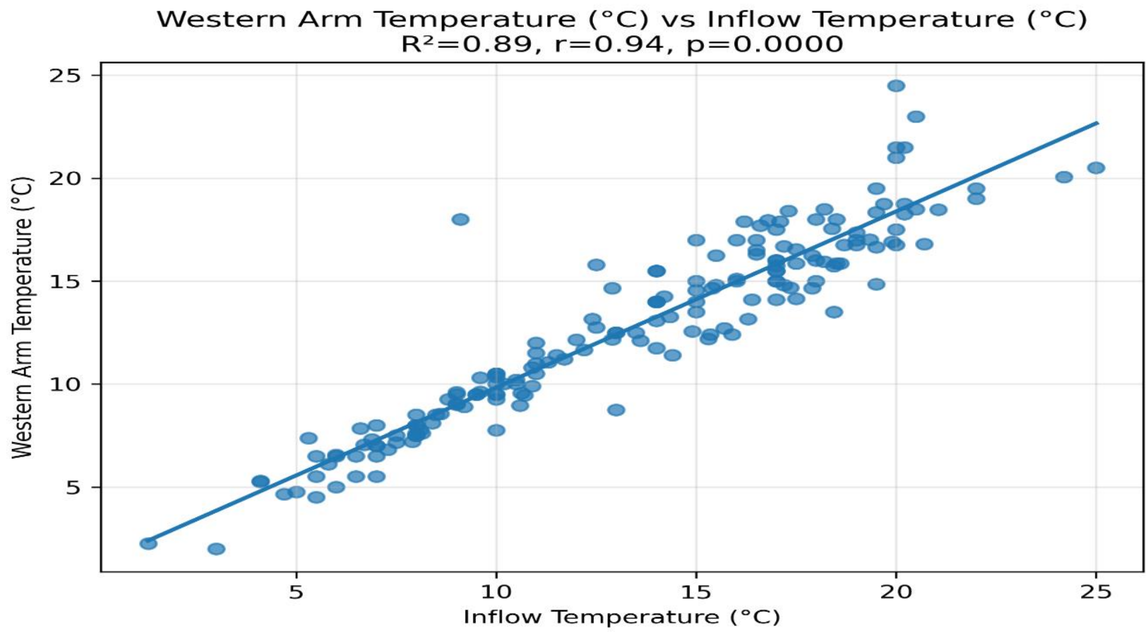

Figure 18. *A scatter plot and a regression line showing a positive relationship between water temperature in the western arm and the water temperature in the western spring inflow. There is a positive relationship between the Western Arm pH and the Western Spring Inflow.*

<u>Comparisons with other data sets.</u>

Comparisons with any other water-quality variables or climatic and water-level datasets. (This could include other variables determined at Stackpole).

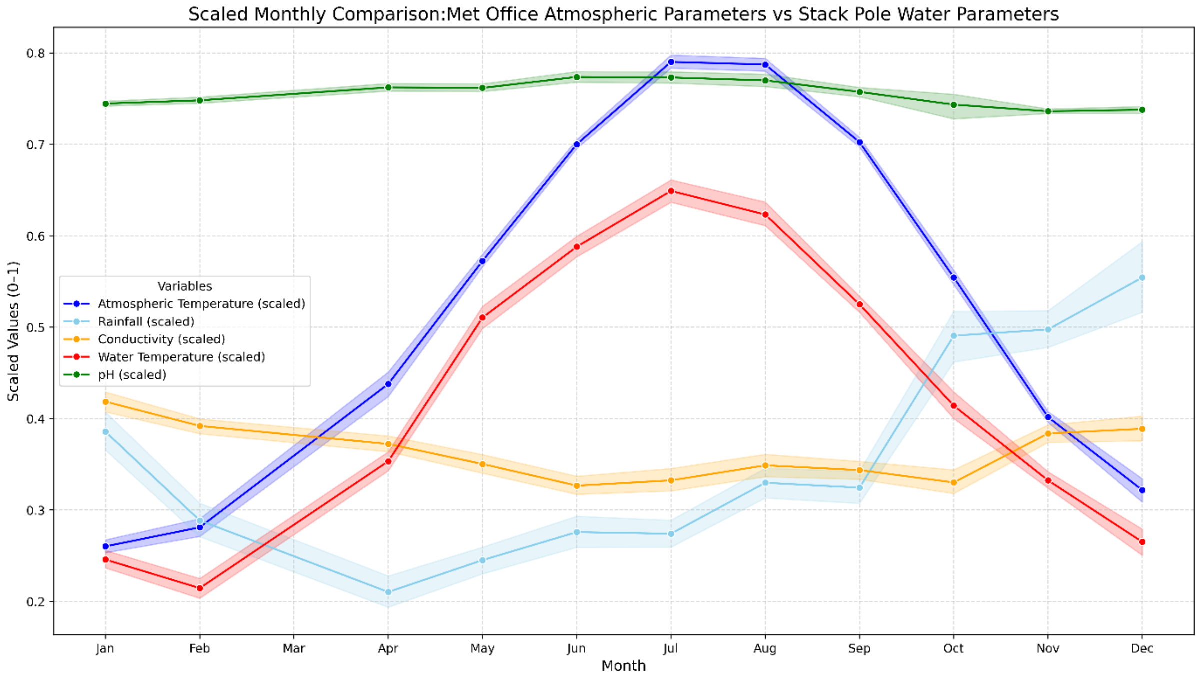

Figure 19. *This figure compares the seasonal patterns of atmospheric and water chemistry variables after scaling each dataset to a 0–1 range. Scaling was necessary because the variables were originally recorded in different units and magnitudes (e.g., °C for temperature, µS cm^{-1} for conductivity, mm for rainfall, and unitless pH). Standardizing them to a common scale allows the shape of seasonal trends to be compared directly.*

Atmospheric temperature data were obtained from the Met Office "Historic Station Data" (Aberporth), which is the nearest long-term monitoring station to Bosherston Lily Ponds (Met Office, https://www.metoffice.gov.uk/research/climate/maps-and-data/historic-station-data). Water temperature closely follows this atmospheric pattern, with both variables rising steadily from winter to a peak in July–August, then decreasing towards winter. This indicates that water temperature is strongly controlled by seasonal climatic forcing.

pH also increases during summer, mirroring the temperature trend. This suggests that higher temperatures promote increased photosynthetic activity, which removes CO_2 from the water and raises pH through shifts in the carbonate equilibrium.

In contrast, conductivity demonstrates an opposite seasonal pattern, decreasing during summer and increasing in winter. This is consistent with greater biological uptake and carbonate precipitation during warmer months, which reduces dissolved ion concentrations in the water. Meanwhile, rainfall peaks in autumn and winter, contributing to dilution and hydrological flushing, reinforcing the lower conductivity observed during high-runoff periods.

Atmospheric yearly temperature and water temperature.

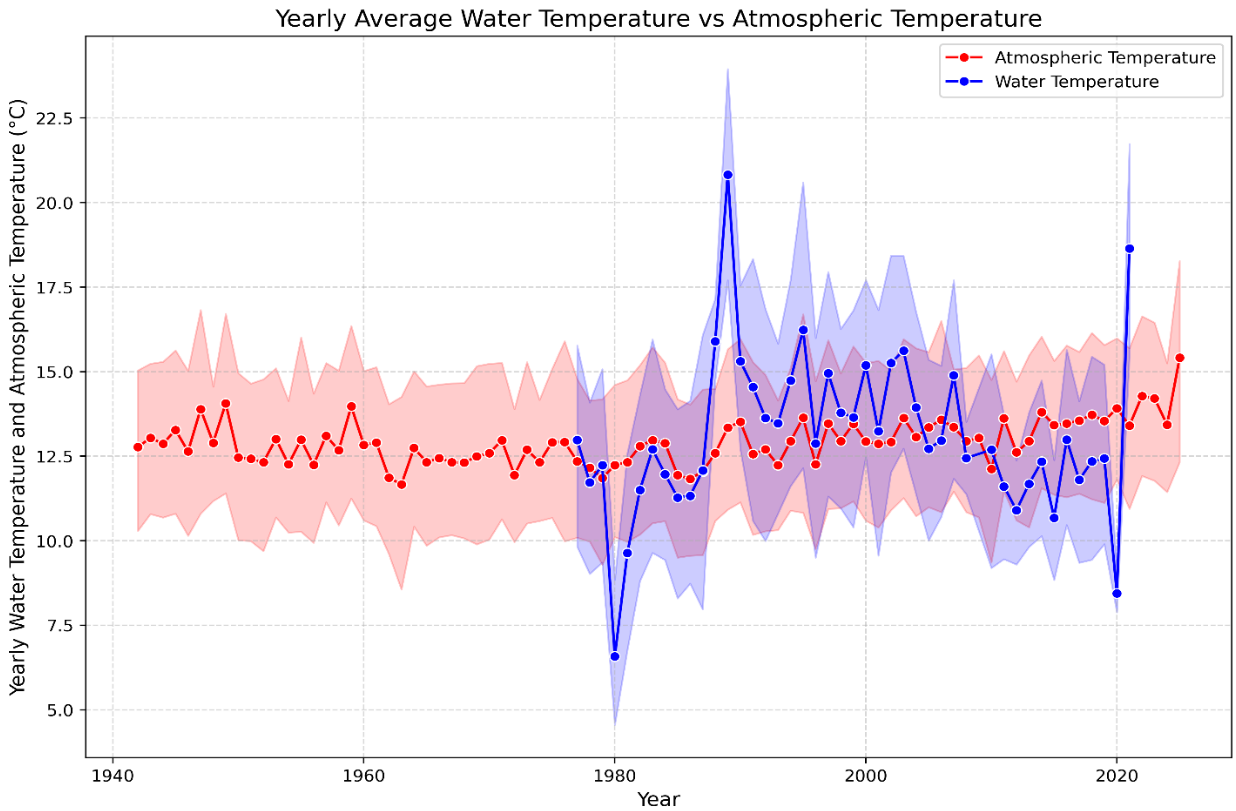

Figure 20. *This graph compares the yearly average water temperature in the Bosherston Lily Ponds system to yearly atmospheric temperature recorded at the Met Office Aberporth station, the nearest long-term climate monitoring site. Water temperature values are shown in blue, and atmospheric temperature in red.*

<u>Quadratic Regression analysis between Atmospheric temperature and Water Temperature.</u>

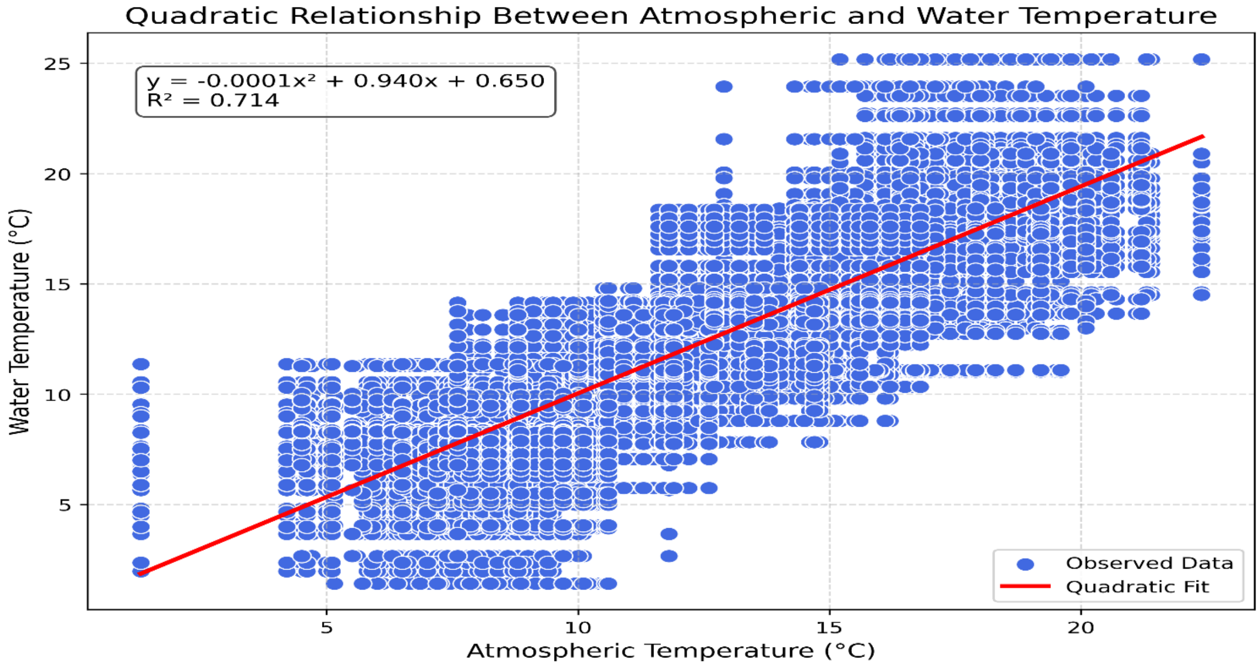

Figure 21. *The scatter plot shows a strong positive relationship between atmospheric temperature and water temperature across the study period. As atmospheric temperature increases, water temperature also increases, although the relationship is not perfectly linear. The fitted quadratic regression model explains this pattern well, with an R^2 value of 0.714, meaning that approximately 71% of the variation in water temperature can be explained by atmospheric temperature alone.*

MACHINE LEARNING MODELLING pH target

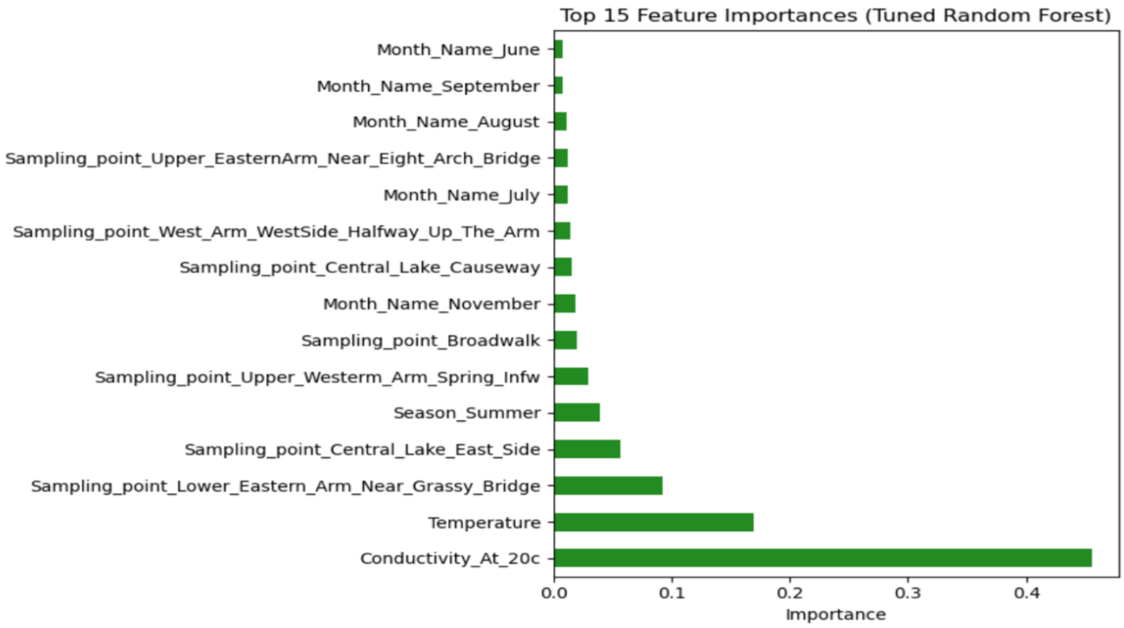

Figure 22. The *Random Forest model identifying the key predictors of pH shows that Conductivity (at 20°C) is the most influential variable in determining pH across the pond system, with a markedly higher importance score than all other predictors. This reflects the strong inverse chemical relationship between pH and ionic concentration: as photosynthetic CO_2 uptake increases during warmer periods, carbonate precipitation removes ions from solution, raising pH and lowering conductivity.*

Interpretation of Partial Dependence between Conductivity and pH

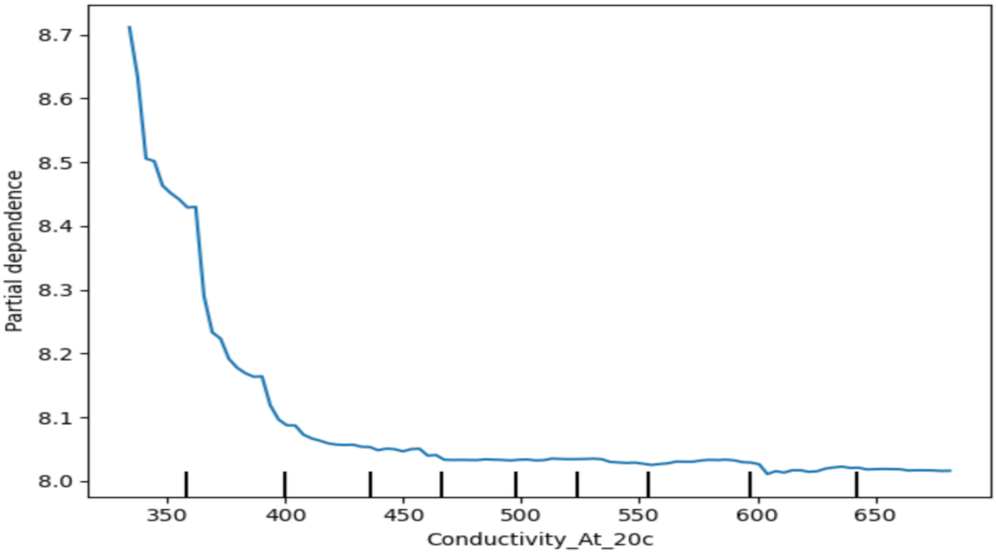

Figure 23. *The partial dependence plot shows the isolated effect of conductivity on pH, while holding all other predictors constant. The curve demonstrates a strong negative relationship: as conductivity increases, pH decreases. At lower conductivity values (~330–380 µS/cm), pH levels are relatively high (≈ 8.6–8.7). However, as conductivity rises above 400 µS/cm, pH gradually declines and stabilizes around 8.0 at higher conductivity levels (> 500 µS/cm).*

Interpretation of Partial Dependence between Temperature and pH

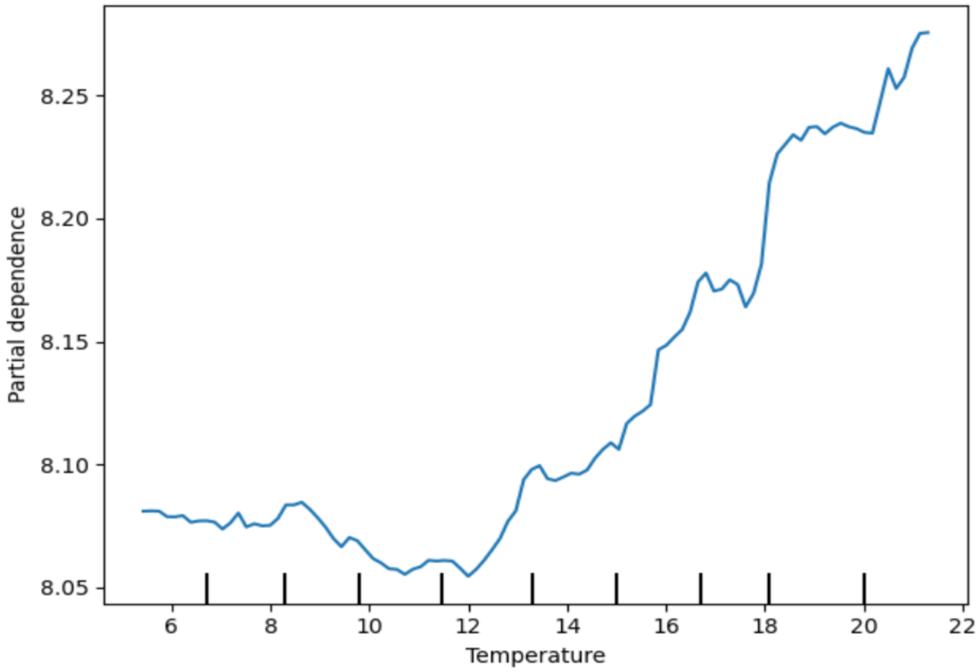

Figure 24. *The partial dependence plot shows the isolated effect of temperature on pH, while holding all other predictors constant. At lower temperatures (approximately 5–12°C), pH values remain relatively stable around 8.05–8.10, indicating minimal temperature influence in colder conditions. However, pH begins to increase steadily once temperatures exceed ~14°C, with the strongest increase occurring between 16°C and 20°C, reaching values of approximately 8.25–8.30 at the warmest observed temperatures.*

Conclusion

This study shows that water temperature in the Bosherston Lily Ponds is strongly controlled by atmospheric temperature, reflecting typical thermal behaviour of shallow freshwater systems. This indicates that the system is highly sensitive to wider climatic warming trends, which has long-term implications for dissolved oxygen availability and the ecological functioning of the ponds. The strong coupling between atmospheric temperature and water temperature observed in this study is consistent with global patterns of lake warming (O'Reilly et al., 2015; Woolway et al., 2021). This has important ecological consequences, as warmer water holds less dissolved oxygen, which increases physiological stress for aquatic organisms (Woolway, Sharma and Smol, 2022; Kraemer et al., 2021).

Water chemistry, particularly pH and conductivity, is strongly influenced by both biological processes and catchment-derived inputs. Increased photosynthetic activity during warmer periods elevates pH, while nutrient-enriched inflow from the Western Arm Spring leads to elevated conductivity and altered chemical dynamics in the Western Arm relative to the other ponds. The historical decline of *Chara* appears consistent with nutrient enrichment causing increased algal biomass and reduced light penetration, reinforcing the importance of nutrient control in maintaining macrophyte-dominated states.

Overall, the Bosherston system displays a combination of climate-driven thermal variation and localised anthropogenic nutrient pressures, resulting in spatial divergence in water chemistry. Management efforts should prioritise reducing nutrient inputs from the surrounding catchment, particularly potassium and other fertiliser-derived ions, while continued monitoring is needed to assess long-term ecological responses. Further research on nutrient export to the adjacent marine environment is recommended to understand downstream impacts.

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   Ndlovu, P.A. (2025) *Bosherston Lily Ponds Water Chemistry Interactive Dashboard*. Available at: https://lkylot2tfqjksmflvu4nfs.streamlit.app/

3. **Jupyter Book**
   Ndlovu, P.A. (2025) *Bosherston Lily Ponds Water Chemistry Analysis – Jupyter Notebook Workflow*. Available at:

Main data analysis
https://kabila96.github.io/data_analysis_bosherstone_lakes_ph_temperature_conductivity_data_analysis_ML_jupyternotebeook/

Atmospheric Temperature vs Water Temperature

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AI assistance (ChatGPT, GPT-5) was used only to rephrase and clarify my own explanations, and to assist with formatting and caption wording. No content was copied directly without modification, and all analysis and interpretations were produced independently.